Incident Report: Insider Threat – Unauthorized Data Access and Exfiltration

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Alert Title

Unusual High Volume Data Access by Internal User

Summary

An internal user `imran.saleem` from the HR department accessed a significantly high number of sensitive files within a short time window outside of normal working hours. The activity involved confidential employee data, including salary records, followed by copying files to an external USB device. This behavior deviates from the user's normal activity baseline and indicates potential insider threat and data exfiltration.

Initial Findings

- User: `imran.saleem`
- Department: HR
- Time: 06:30 PM – 07:15 PM
- Files Accessed: 200+
- File Types: Employee records, salary sheets, confidential documents
- Activity: Bulk access + USB transfer
- Privileges: Standard user (no admin rights)

Key Observations

- Access outside standard working hours
- Unusually high volume of file access
- Sensitive HR data targeted
- Data copied to external USB device
- No prior history of similar activity
- Behavior deviates from baseline user profile

Investigation

File Access Activity

```
DeviceFileEvents
| where AccountName == "imran.saleem"
| summarize FileAccessCount = count()
    by DeviceName, bin(TimeGenerated, 15m)
| where FileAccessCount > 100
| sort by FileAccessCount desc
```

USB Data Transfer Detection

```
DeviceFileEvents
| where AccountName == "imran.saleem"
| where FolderPath contains "USB" or FolderPath contains "Removable"
| summarize FileCopied = count() by DeviceName, bin(TimeGenerated, 15m)
| sort by FileCopied desc
```

Result: Confirmed abnormal spike in file access and data transfer to removable media.

Evidence

- 200+ files accessed within ~45 minutes
- Sensitive HR documents targeted
- Data copied to external USB device
- Activity occurred outside normal working hours
- No similar historical behavior

Conclusion

This incident is classified as a **Confirmed True Positive (Insider Threat – Data Exfiltration)**. The user performed unauthorized bulk access of sensitive data followed by transfer to external media, indicating potential data theft.

Severity

Critical

Response Actions

- User account temporarily suspended
- Endpoint isolated for forensic investigation
- USB access disabled across environment (as precaution)
- Data Loss Prevention (DLP) rules reviewed and enforced
- Incident escalated to Tier 2 / Security Management
- HR and compliance teams notified

Analyst Note

This incident highlights the importance of implementing strict Data Loss Prevention (DLP) controls and monitoring user behavior baselines. Organizations should enforce policies for removable media usage and trigger alerts on abnormal file access patterns, especially involving sensitive data.